Terminology, Formula Interpretation, and Experimental Methodology of the DeReFusion Codebase

Applied Soft Computing Study Project

Based on the repository snapshot `Alastair-Jiang/DeReFusion`, branch `main`, commit `87bc113` (2026-10-13)

***Abstract*—This report consolidates and rewrites two earlier terminology documents. It is addressed to readers who must explain and defend the project, and it provides three classes of content. First, definitions of the terms used by the project, together with the exact location of each term in the codebase. Second, an interpretation of several key formulae found in the code, in particular the four groups covering decomposition, normalization, the residual branch, and the fusion operator, with attention to the relation between their mathematical meaning and their implementation. Third, a complete account at the methodological level, comprising the evaluation and statistical decision system, the full evidence from the ablation experiments, and the governance apparatus formed by the L0–L5 gates, execution releases, and receipts. Known inconsistencies of wording and gaps in the evidence are stated alongside. It must be emphasized that this is a terminology and methodology document, not an experimental report, and that it constitutes neither publication of results nor any authorization of data access or execution.**

***Index Terms*—Financial time-series forecasting; decomposition–residual architecture; reversible instance normalization; fusion strategy; controlled comparison; pre-registration**

I. Background and Scope

A. Three entities that must be distinguished

When discussing this project, the following three entities must be described separately; conflating them leads to errors of citation and of attribution.

The original paper. Chih-Chien Hsieh and Mu-Yen Chen (corresponding author), Department of Engineering Science, National Cheng Kung University, Taiwan, published in *Applied Soft Computing*, vol. 203 (2026), article 116252, DOI `10.1016/j.asoc.2026.116252`. The method, the original experimental design, and the paper's conclusions are the work of those authors.

This repository. `Alastair-Jiang/DeReFusion`, an *independent research branch*. Its own README states explicitly that the project is an independent research branch built on the public DeReFusion code and on THUML Time-Series-Library, and that it is *not the official code package released by the paper's authors*. The code skeleton follows THUML Time-Series-Library, which accounts for the `run.py`, `exp/` and `data_provider/` directory layout.

Local artifacts. The 174 entries under the local `results/` directory and the 175 under `checkpoints/`, covering 47 assets and produced on CPU, are original artifacts unique to this machine and are not distributed with the remote repository.

B. Sources and snapshot

All factual statements in this report derive from commit `87bc113` of the repository's `main` branch (merged 2026-10-13), whose README carries a report update dated 2026-10-10; figures on the paper side are taken from the original PDF. Should the repository be updated subsequently, its most recent state governs.

C. Nature and boundaries of this document

This is a document about terminology and methodology, not an experimental report. It publishes no datasets, prediction arrays, or weights, and it constitutes no authorization for access to real data, for preprocessing, for fitting, or for scoring. Interpretations of experimental results in the repository must remain bound to their own protocol, manifest, date, and attempt identity; a newer filename does not supersede an earlier approval or pre-registration. Every reference to a result in this report retains the evidence grade assigned in the original report and is not upgraded.

II. Problem Setting, Data, and Splitting

A. Forecasting task

OHLC. Open, High, Low, and Close. The model reads four fields (`enc_in=4`, `dec_in=4`) and forecasts only the close (`c_out=1`, argument `--target Close`). The data reside in `dataset/` and `data/raw/yahoo/`; the registry `reproduction/results/dataset_registry.csv` lists 61 daily-frequency CSV files totalling 149 474 rows.

Time-series forecasting and long-term forecasting. "Long-term" here does not mean a distant horizon but rather the prediction of several future steps at once. The entry point is `run.py --task_name long_term_forecast`, and the training and evaluation loop is in `exp/exp_long_term_forecasting.py`. The paper covers $T \in \{1, 7, 12, 24, 36\}$; Phase 1 of the repository narrows this to $\{1, 24\}$.

Window parameters. `seq_len` is the input window length, `pred_len` the forecast length, and `label_len` the history consumed by the decoder. This model uses no decoder: the forward function of `DeReFusion` ignores `x_dec`, `x_mark_dec`, and `mask` outright and reads only `x_enc`. The default configuration is `seq_len=96`, `label_len=48`, `pred_len=24`.

Assets and grouping. Phase 1 comprises 30 assets in two confirmatory strata, `original-10` and `c1-20`. The two strata

must be reported separately in any report; a single pooled mean may not be used to conceal the difference between them.

Non-stationarity. The paper reports that KPSS tests reject stationarity at the 1% level for all ten assets; the series exhibit volatility clustering and a Hurst exponent near 0.5, close to a random walk. This is the background to the finding that longer look-back windows increase error.

B. Splitting and the division of labour between the two normalizations

Chronological splitting. Time series must be split in chronological order and never shuffled. The paper splits 70%/10%/20% in order; the repository's pre-registration offers two modes, by ratio or by date (`--split_mode dates` together with `--train_end`, `--val_end` and `--test_end`, with end-exclusive boundaries). The audit script `reproduction/analysis/audit_phase1_date_splits.py` checks all 72 asset/year/horizon combinations.

Data leakage. The entry of future information into training or model selection, which inflates reported metrics. The usual forms are shuffling before splitting and fitting normalization parameters on the full sample. The frozen design requires chronological splitting and fits the scaler on training rows only.

Z-score standardization versus RevIN. The two operate at different scales. Z-score standardization rescales the whole series once, using training-segment mean and standard deviation, and is implemented in the data loader under `data_provider/`. RevIN rescales each input window using that window's own mean and standard deviation and inverts the transform after prediction; it is implemented in `layers/RevIN.py`. Doctrine clause L4 requires all dynamic transforms to be fitted on the training prefix only, which addresses precisely this class of leakage.

C. Stated policy on data quality

Known defects. Two classes of problem are recorded publicly rather than silently repaired: first, five foreign-exchange series violate the OHLC envelope (with a high below a low, for instance); second, WTI crude retains non-positive prices from April 2020. The rule is to keep the raw data and to refrain from adding constants or deleting rows to conceal the issue; metrics involving logarithms must mark those windows as undefined rather than altering prices. The relevant sources are Section 2 of the pre-registration, Section 5 of the README, and `reports/evidence_closure/15_data_acquisition_diagnosis.md`.

Calibration-only assets. The three assets `BOND10Y`, `BOND ETF` and `CITICSEC` are used only to calibrate the pipeline and may never enter any Phase 1 result.

III. Model Architecture and Formula Interpretation

This section states the formulae that map directly onto the implementation, and explains the relation between their mathematical meaning and their implementation details. All formulae are written from the actual code on the repository's `main` branch.

A. Overall data flow

DeReFusion consists of two parallel branches wrapped in reversible instance normalization, whose outputs are added directly:

$$\hat{Y} \;=\; \mathrm{RevIN}^{-1}\big(\, Y_{\text{base}} \;+\; Y_{\text{res}} \,\big), \tag{1}$$

where Y_{base} is the output of the DLinear base branch and Y_{res} that of the nonlinear residual branch; both have shape $[B,\, T,\, C]$, with B the batch size, T the forecast length, and C the number of channels. The addition in (1) carries no learnable parameter.

a) Key point: The object under comparison in this project is therefore not either branch but the fusion operator itself. The original paper likewise positions DeReFusion as a controlled testbed: architecture, data, and protocol are fixed and only the fusion operator varies.

B. Moving-average decomposition

The trend component is a moving average with edge replication, and the seasonal component is defined as the residual after subtracting it:

$$X_{\text{tr}}[t] = \frac{1}{k}\sum_{i=-p}^{p} X_{\text{pad}}[t+i], \qquad p = \left\lfloor \frac{k-1}{2} \right\rfloor, \tag{2}$$

$$X_{\text{se}} \;=\; X - X_{\text{tr}}. \tag{3}$$

The implementation is `MovingAvg` and `SeriesDecomposition` in `layers/derefusion_components.py`: the former uses `nn.AvgPool1d` with replication at both ends, and the latter returns `(x-trend,trend)`. The default kernel width is $k = 25$, read through `getattr(configs,"moving_avg",25)`.

a) Key point: The moving average is a low-pass filter, not a proof of trend. The kernel width k defines what counts as slow: variation inside the window falls to the seasonal component, variation beyond it to the trend. The decomposition is therefore **scale-relative**, and changing k changes the boundary between the two components; $k = 25$ (about one trading month) is a tunable modelling choice, not a natural constant. Moreover, the "seasonal" component in (3) carries no periodic meaning; it is simply the complement of the trend.

C. The DLinear base branch

The two components pass through separate linear maps and are then summed:

$$Y_{\text{base}} \;=\; W_{\text{se}}^{\top} X_{\text{se}} \;+\; W_{\text{tr}}^{\top} X_{\text{tr}}, \qquad W_{\text{se}}, W_{\text{tr}} \in \mathbb{R}^{L\times T}, \tag{4}$$

where L is the input length and T the forecast length. The implementation is `DLinearBranch`: each component is permuted to $[B, C, L]$, passed through its own `nn.Linear(seq_len,pred_len)`, and the results are summed and permuted back to $[B, T, C]$.

a) Key point: Separating W_{se} from W_{tr} gives each time scale its own map, so that the slow evolution of the trend and the periodic oscillation of the seasonal component need not share weights; this structurally reduces interference between modes. An easily overlooked implementation detail should be noted at the same time: `DLinearBranch` uses **channel-shared** linear maps and has no per-channel (`individual`) switch, whereas the principal opponent `revin-DLinear` supports channel-independent mode and initializes its linear weights to the constant $1/L$, while the DeReFusion linear branch retains PyTorch default initialization. That difference in initialization is shown by the ablation experiments to matter materially (see Section VI).

D. Reversible instance normalization

Writing $\mu(x)$ and $\sigma(x)$ for the mean and standard deviation of a window x along the time axis, normalization and denormalization are

$$\tilde{x} = \frac{x - \mu(x)}{\sqrt{\sigma^2(x) + \epsilon}} \cdot \gamma + \beta, \tag{5}$$

$$\hat{y} = \Big(\tilde{y} - \beta\Big)/\gamma \cdot \sqrt{\sigma^2(x) + \epsilon} + \mu(x). \tag{6}$$

The implementation is `layers/RevIN.py`: the mean and standard deviation are computed along the time axis with shape $(B, 1, C)$ and are `detach`ed, then stored on the module, and denormalization reuses the same statistics. The default is `affine=True`.

a) Key point: The precise meaning of "reversible" is that (6) reuses the $\mu(x)$ and $\sigma(x)$ stored during (5), and does **not** recompute statistics on the forecast. This design lets the model operate in a common scale while returning output in the original price scale, and it is what makes comparison across assets possible. Every opponent model carries the same wrapper (hence the uniform `revin-` filename prefix), which is a component of the controlled comparison.

E. The nonlinear residual branch

The residual branch is a cascade whose composition order is

$$Y_{\text{res}} \;=\; W_o\Big(W_\tau\big(\text{Trans}(\text{LSTM}(W_e\tilde{x}))\big)\Big), \tag{7}$$

where $W_e : \mathbb{R}^C \to \mathbb{R}^d$ is the feature projection, the LSTM has a single layer with hidden width d, `Trans` is a single-layer Transformer encoder (H heads, feed-forward width $4d$, dropout 0.3), $W_\tau : \mathbb{R}^L \to \mathbb{R}^T$ acts on the time axis, and $W_o : \mathbb{R}^d \to \mathbb{R}^C$ is the channel projection.

a) Key point: two senses of the word "residual": This is the point most easily misstated, and the two senses must be kept apart. As an *implementation*, the residual branch is a prediction head that runs in parallel with the linear branch and is parameterized independently: it learns its own parameters, emits a tensor of the same shape as Y_{base}, and is then added to it. Its nature is that of two forecasters being summed, not that of a residual fit in the mathematical sense. As an *analysis*, Section 4.5.3 of the paper defines the residual $r = y - \hat{y}_{\text{lin}}$ (what the linear branch leaves unexplained) and the net contribution of the nonlinear branch $\hat{r} = \hat{y}_{\text{full}} - \hat{y}_{\text{lin}}$, in order to test whether the latter is learning the former. The two are not interchangeable: the one is model structure, the other a post hoc attribution analysis. In addition, the module docstring calls the branch "TransformerLSTM" although the execution order is **LSTM first, Transformer second**; citation should follow (7).

F. Fusion operator and gating variants

Parameter-free additive fusion is (1). The three gating variants share one form:

$$g = \sigma\big(A(x) + b_0\big), \qquad Y = (1 - g) \odot Y_{\text{base}} + g \odot Y_{\text{res}}, \tag{8}$$

where $(1 - g)$ and g form convex weights, b_0 is a bias parameter initialized to -2.0, and σ is the sigmoid function. The variants differ only in $A(\cdot)$:

- **v1, volatility-aware** (`DeReFusion-gatev1-volatilityaware.py`): A takes the volatility of the *raw input*, $\text{std}(x_{\text{raw}}) + 10^{-5}$, of shape $[B, 1, C]$, expanded to $[B, T, C]$ by `Linear(1, pred_len)`. *In this repository:* this path compresses the volatility of the entire window into one scalar per channel, a pronounced information bottleneck.
- **v2, learnable** (`DeReFusion-gatev2-learnable.py`): $A \equiv 0$ and $g = \sigma(\alpha)$, with α a parameter of shape $[1, 1, C]$ initialized to -2.0. *In this repository:* the gate is **entirely independent of the input** and returns the same weights for every sample and every time step.
- **v3, input-conditioned** (`DeReFusion-gatev3-inputconditioned.py`): $A(\tilde{x}) = W_2\,\text{ReLU}(W_1\tilde{x})$, with $W_1 \in \mathbb{R}^{L \times m}$ and $W_2 \in \mathbb{R}^{m \times T}$, bottleneck $m = 4$. *In this repository:* this is the only one of the three that is genuinely per-sample, per-channel and per-timestep conditional.

a) Key point: Equation (8) yields three inferences readable directly from the formula. First, the bias initialization $b_0 = -2$ corresponds to an initial gate value $g = \sigma(-2) \approx 0.119$, so that at the start of training roughly 88% of the weight rests on the base branch; the paper's measured mean gate values of 0.123 to 0.140 are nearly identical to that initial value, indicating that the gate is **barely activated** during training. Second, the v2 gate value contains no input variable, and is mathematically equivalent to attaching a learnable global scalar weight to the addition; its per-sample adaptivity is identically zero, so "learnable gating" degenerates here into a constant. Third, (8) is a convex combination, so the gate can in structure only redistribute weight between the two branches and cannot express any combination beyond addition. All three are consistent with the paper's finding that direct addition wins in 45 of 50 asset-by-horizon settings.

G. PyTorch as the implementation substrate

The project's principal language and framework are Python and PyTorch. All models are defined under `models/` as subclasses of `nn.Module`: linear layers as `nn.Linear`, recurrent

networks as `nn.LSTM`, attention as `nn.TransformerEncoder`, and the loss as `nn.MSELoss()`. The training and evaluation loop is in `exp/exp_long_term_forecasting.py` and the unified entry point is `run.py`; dependencies are grouped into three files under `requirements/` (core, CUDA, and optional foundation models).

Inductive bias. The structural assumption a model makes a priori. DLinear assumes that the series can be extrapolated roughly linearly; a Transformer assumes that any two time points may be related. One of the paper's central conclusions is that its advantage comes from an inductive bias matched to the structure of the series rather than from added capacity.

Neural network. A model family built from stacked neurons and fitted by backpropagation and gradient descent. The training details of this project are: MSE loss, Adam optimizer, learning rate 10^{-4}, batch size 32, at most 30 epochs, and early stopping on the validation set with patience 5. The residual branch is the concrete form that family takes here, and its necessity is examined by the ablation experiments.

Frozen training protocol. The pre-registration fixes a common training configuration: `seq_len=96`, `label_len=48`, `enc_in=4`, `dec_in=4`, `c_out=1`; `d_model=32`; moving-average kernel 25; Transformer with 8 heads, two encoder layers, one decoder layer, `d_ff=2048`, dropout 0.1; Adam, learning rate 10^{-4}, MSE loss, cosine schedule; at most 30 epochs, patience 5, validation-selected checkpoint; and explicitly no model-specific hyperparameter search.

Random seeds. `run.py` exposes two distinct arguments: `--rand_seed` (default 2021, used to seed the random generators) and `--seed` (default 2, used only for data augmentation). The pre-registration assigns seeds as follows: $\{2021\}$ for calibration and screening, $\{2022, 2023, 2024\}$ for the primary paired comparison, and $\{2024\}$ for temporal robustness. The screening seed 2021 is excluded from the primary estimate because its results were observed.

Early stopping. Training stops when the validation metric fails to improve for five consecutive epochs; the best validation checkpoint is retained, up to a maximum of 30 epochs.

IV. The Opponent Model System

A. Meaning of the `revin-` prefix

The ten files `models/revin-*.py` all carry this prefix, indicating that the same backbone is uniformly wrapped in the same reversible instance normalization as DeReFusion. This is part of the controlled comparison and removes differences attributable to normalization.

B. Models actually run in Phase 1

The pre-registered model list contains five entries: `revin-DLinear`, `DeReFusion`, `revin-PatchTST`, `revin-iTransformer`, and `revin-TimesNet`.

revin-DLinear. The principal opponent. A linear decomposition model, $Y = W_{\text{se}}^{\top} X_{\text{se}} + W_{\text{tr}}^{\top} X_{\text{tr}}$, with weights initialized to $1/L$. Stage C of Phase 1 is precisely the paired comparison of DeReFusion against this model.

revin-PatchTST. Splits the series into patches used as tokens and models each channel independently.

revin-iTransformer. Treats the variate dimension as tokens, with attention applied across channels.

revin-TimesNet. Folds the one-dimensional series into a two-dimensional periodic tensor and extracts periodic structure with two-dimensional convolutions. The paper reports 2355.5k parameters, about 90 times those of DeReFusion, at comparable accuracy.

C. The benchmark family in the paper

The paper compares ten benchmarks whose model definitions are all present in the repository but which *are not* part of the Phase 1 run list: DLinear, LightTS, ETSformer, PatchTST, iTransformer, Informer, Reformer, FEDformer, Autoformer, and TimesNet. The reported average ranks are ETSformer 3.12, DeReFusion 3.18, TimesNet 3.20, and PatchTST 3.58, which are statistically indistinguishable; heavy attention models such as Informer and Reformer are significantly outperformed by DeReFusion (Friedman $\chi^2 = 357.08$, $p < 0.001$; Nemenyi critical difference $\text{CD} = 2.14$).

D. Zero-shot foundation models and declared gaps

Zero-shot foundation models. `models/Chronos.py`, `models/Moirai.py`, and `models/TimesFM.py` require the optional dependency set in `requirements/foundation.txt`. They forecast without any training and therefore do not belong to the same comparison system as the models above, each of which is retrained per asset; they must not be merged into one ranking. The paper reports better statistical accuracy for all three.

Declared gaps. The pre-registration lists `declared_gaps`: TimeMixer and a state-space model (Mamba). `layers/MambaBlock.py` exists but is not wired into the comparison.

Channel sharing and channel independence. OHLC has four channels. Channel sharing means that all four channels use one set of linear weights; channel independence means one set per channel. DeReFusion uses the former; `revin-DLinear` supports the latter.

V. Evaluation Metrics and Statistical Decision

A. Metrics

Error metrics. MSE, MAE, RMSE, R^2, MAPE. Both the primary metric and the loss are MSE (`nn.MSELoss()`, see `exp/exp_long_term_forecasting.py`). At $T = 1$ the paper reports MSE 0.037 and R^2 0.948. MSE values from different assets or horizons must not be compared directly without qualification.

Directional accuracy. Whether the direction of movement is predicted correctly, against a chance baseline of 50%. The paper reports 50.82%.

Sharpe ratio and buy-and-hold. The Sharpe ratio measures return per unit of volatility and is a risk-adjusted efficiency measure; buy-and-hold denotes holding the asset without trading. The paper reports a Sharpe ratio of 0.26 at zero cost, falling to -0.35 once 10 basis points of transaction cost are included, against a buy-and-hold Sharpe of 0.62 and annualized return of 17.77%.

B. Paired comparison and stratification

The primary comparison quantity is $\Delta\text{MSE} = \text{MSE}(\text{DeReFusion}) - \text{MSE}(\text{revin-DLinear})$, negative values indicating lower error for DeReFusion. It is reported separately by stratum, horizon, and seed, giving 12 stratified point estimates. The current state: at $T = 1$ every stratum is negative; at $T = 24$ some cross zero or turn positive (for instance `original-10`, $T = 24$, seed 2023 at $+0.004682$ and seed 2024 at $+0.000922$).

Strata. `original-10` and `c1-20` must be reported separately in any report. The pre-registration states: "no single pooled mean may hide the two strata."

C. The family of statistical tests

Friedman test. An overall rank test for many models across many settings. The paper reports $\chi^2 = 357.08$, $p < 0.001$, indicating systematic differences among models.

Nemenyi test. A post hoc pairwise procedure following Friedman, yielding the critical difference in average ranks; the paper uses $\text{CD} = 2.14$ to determine which models are statistically indistinguishable.

Wilcoxon signed-rank test. A paired nonparametric test comparing the distribution of differences for two models on the same pairing.

Diebold–Mariano test. The standard tool of the forecasting literature for comparing the errors of two forecast series.

All three are used together here so that conclusions do not depend on the idiosyncrasy of any one method.

D. Multiplicity, selection overfitting, and trial count

Holm correction. Tightening the significance threshold when several pairwise comparisons are made within one family, to offset chance significance. The pre-registration promises this correction within the modern-model family at Stage B; an earlier review found no corresponding implementation in the repository, and the point should be re-verified before citation.

Selection overfitting, PBO, and CSCV. When the best performer is selected from many candidates, part of its apparent superiority is luck. PBO (probability of backtest overfitting) quantifies that component and CSCV (combinatorially symmetric cross-validation) is the estimator. Doctrine clause L4.8 states: "a value above 0.05 is a rejection, not a caveat."

Trial count N. Principle P4 requires that N be fixed and declared before the first run, with each seed counted as a trial, and that it be reported honestly afterwards with a deflation table for the winner. A result without its N is "not interpretable." The same document acknowledges a gap: the deflation machinery is derived for a Sharpe-like selection criterion and does not transfer to an error-metric protocol, so the trial count currently stands as an uncorrected caveat.

Bootstrap confidence intervals. The pre-registration requires asset-cluster bootstrap intervals with an immutable seed and equal asset weighting. However, the seed value, the number of resamples, the interval algorithm, and the pooling rule across horizons and seeds were never frozen in advance, so the confirmatory interval cannot currently be computed and the Stage C conclusions remain descriptive only.

E. Pre-registration and the three permitted claims

Pre-registration and protocol identifier. The document is `docs/PHASE1_PREREGISTRATION.md`, protocol `phase1-v1.1-2026-09-19`, frozen before any Phase 1 outcome existed.

The three permitted claims (verbatim). The study may support only one of the following claims:

1) **consistent benefit**: DeReFusion has a negative paired loss difference relative to DLinear in both frozen asset strata, with a cluster-bootstrap 95% interval excluding zero and without a cost increase that dominates the accuracy gain;
2) **conditional benefit**: the pooled interval or one stratum crosses zero, but pre-declared asset and horizon summaries show repeatable heterogeneity;
3) **no demonstrated benefit**: neither condition is met.

The third is a legitimate conclusion and stating it does not constitute failure. It is followed immediately by a prohibition: no outcome licenses an asset-level router, an $|\text{ACF1}|$ rescue, or a physical or Navier–Stokes mechanism claim.

Prohibited actions. No post hoc feature search; no claim that higher volatility implies a preferred nonlinear operator; no revival of sample-level gating, routing, or mixture-of-experts; no use of test loss or manual model selection during training; no single pooled score concealing failures at the asset, seed, or horizon level; no use of "physics-informed," "conservation," or "Navier–Stokes" without a specified, measured physical constraint; and no removal of an asset because its sign is inconvenient.

VI. The Full Process of the Ablation Experiments

A. Definition and decision logic

An ablation removes one component of a model at a time and observes the change in performance: substantial degradation indicates that the component is effective, no change indicates no material contribution, and improvement indicates a negative contribution. The logical premise is that everything else is held fixed. It must be stressed that executing an ablation and establishing the necessity of a component are two different things, and the evidence matrix of this project labels them separately.

B. Ablation design in the paper

The paper's ablation comprises four variants: the full model, the removal of the LSTM, the removal of the Transformer, and the removal of the entire nonlinear residual branch (leaving only the DLinear base branch). Apart from the removed component, the backbone, preprocessing, splitting, look-back length $L = 96$, the five horizons, the five random seeds, and the hyperparameters are all identical.

Three conclusions follow. First, the nonlinear residual branch contributes most at short horizons: at $T = 1$ the full model attains 0.037, about 27% below the 0.051 of the linear-branch-only variant. Second, the LSTM and the Transformer complement each other, with the Transformer mattering slightly more at short horizons: at $T = 1$, removing the Transformer degrades performance to 0.047, a larger change

TABLE I
PAPER ABLATION RESULTS: MSE (10 ASSETS × 5 SEEDS)

Variant	$T=1$	$T=7$	$T=12$	$T=24$	$T=36$
Full model (DeReFusion)	**0.037**	**0.081**	0.117	0.205	0.295
w/o LSTM	0.041	0.083	**0.116**	**0.197**	**0.284**
w/o Transformer	0.047	0.082	0.116	0.204	0.293
w/o residual branch	0.051	0.093	0.127	0.203	0.285

TABLE II
RESIDUAL COMPLEMENTARITY: $\hat{r}$ AGAINST r

Metric	T=1	T=7	T=12	T=24	T=36
Correlation	0.522	0.349	0.294	0.159	0.099
Error reduction	+27.4%	+12.2%	+8.1%	−0.3%	−3.4%
OLS slope	1.029	0.954	0.798	0.480	0.311

than the 0.041 from removing the LSTM. Third, long horizons belong to the linear base branch, where the residual branch's gain fades and even reverses: from $T \geq 12$ onward the LSTM-removed variant is best, and at $T = 36$ the full model ranks last (0.295, worse than 0.284 without the LSTM and 0.285 without the residual branch).

The paper states: "within the tested architecture, the linear base branch carries most of the long-horizon performance, while the extra nonlinear capacity adds little and can even reduce accuracy. The present ablation does not isolate the mechanism; error accumulation in the recurrent LSTM is one possible contributor." This statement bounds the scope of the conclusion.

On the relation between the residual-branch-removed variant and a standalone DLinear, the paper reports a correlation of 0.999 between their predictions, though the values are not identical (0.051 versus 0.049 at $T = 1$). That small difference proves material on the repository side; see Section VI-E.

C. Residual complementarity analysis

To test whether the nonlinear branch is learning what the linear branch leaves unexplained, the paper defines the actual residual $r = y - \hat{y}_{\text{lin}}$ and the net contribution of the nonlinear branch $\hat{r} = \hat{y}_{\text{full}} - \hat{y}_{\text{lin}}$, computed on the same batch of aligned windows.

a) Key point: At $T = 1$ the correlation is 0.522, the slope 1.029 (nearly unbiased, indicating that the magnitude also matches), and the error reduction 27.4%, showing that the branch is indeed learning residual dynamics beyond the reach of the linear branch rather than supplying redundant capacity. As the horizon grows, however, the correspondence weakens monotonically: by $T \geq 24$ the correlation remains positive while the net error reduction turns negative and the slope falls below 0.5, indicating that the contribution fades and can turn into interference.

D. Controlled experiments on the fusion operator and the look-back length

Fusion operator. With the architecture fixed, only the fusion operator is replaced, across 50 asset-by-horizon settings. Direct addition attains 0.037 at $T = 1$, better than the learnable gate at 0.049, the input-conditioned gate at 0.052, and the volatility-aware gate at 0.054, and it wins 45 of the 50 settings. Increasing complexity is accompanied by increasing degradation (mean MSE 0.147, 0.156, 0.158, 0.159 in that order). The learned gate values remain persistently base-favoring: mean gate values between 0.123 and 0.140, with at most 0.04% of test samples above a gate value of 0.5 and none above 0.9. The gates moreover exhibit **underfitting** rather than overfitting: gate training loss is 0.040 to 0.041 against 0.037 for direct addition, with generalization gaps of 0.018 and 0.019.

Look-back length. For $L \in \{48, 96, 192\}$, the $T = 1$ MSE is 0.036, 0.037, and 0.087 respectively, and the average MSE over the five horizons rises from 0.133 to 0.147 and then to 0.227. The conclusion is that longer look-backs are associated with higher errors, consistent with weak long-term memory. Heavy attention models collapse at $L = 192$, whereas the average MSE of DeReFusion rises by about 71% while R^2 remains above 0.42.

Ablations the paper does not perform. The paper performs no normalization on/off ablation, no decomposition on/off ablation, and no dropout or capacity sweep. The authors state two points explicitly in the limitations: first, that the cascade order (LSTM first) relative to a parallel arrangement or to pure attention "is not ablated here. It is therefore a design choice rather than a verified optimal configuration"; second, that the study deliberately adopts a lightweight and model-agnostic reversible instance normalization and does not include more refined adaptive normalization. These points should be reported faithfully in any external account.

E. Reruns and adjudication on the repository side

The repository reran the ablation independently and adjudicated each item by evidence grade.

The GSPC ordering at $T = 24$ is DeReFusion (0.0623) $<$ DLinear (0.0701) $<$ the gatev2 learnable gate (0.0719).

F. Capacity and routing

Capacity sensitivity. Only the hidden width of the nonlinear operator is varied (23/64/128); all else is identical. A negative ΔMSE indicates that the nonlinear operator is better:

The three assets yield three different conclusions: ETHUSD turns entirely to the nonlinear operator once capacity suffices; GSPC approaches parity as capacity grows; BTCUSD prefers the linear operator at every capacity and the gap widens. The verdict is capacity-sensitive (high for ETHUSD, moderate for

TABLE III
ADJUDICATION OF ABLATIONS AND RELATED CONTROLS ON THE REPOSITORY SIDE

Item	Result and verdict
Component ablation rerun (6 runs, second machine, single seed 2021, identical protocol)	Environment calibration reproduced (0.070065 against 0.07007). Removing the Transformer is 10.71% worse than the full model; removing the LSTM is 1.29% **better** than the full model on GSPC; on BTCUSD all three variants are worse than the full model. Verdict: **partially supported**—component necessity is not established (one seed, two assets).
Validity of the residual-branch-removed ablation	The variant is architecturally identical to `revin-DLinear` (4664 parameters) yet differs by 8.28%, traced to the weight-initialization convention ($1/L$ constant against the PyTorch default). Verdict: **not supported** (`CONFOUNDED_BY_INIT`)—it cannot be read as a component-removal effect.
Cross-machine reproduction of the baseline	An independently configured second machine reproduced the GSPC $T = 24$ `revin-DLinear` calibration (0.070065 against 0.07007). Verdict: **supported**.
Gating control	On GSPC, the v1 gate gives 0.06767 against 0.06230 for direct addition. Verdict: **not supported**.

GSPC, none for BTCUSD). The governing rule requires the more cautious formulation: reading "linear is better" wholesale as a representation bottleneck is not permitted, and the weaker statement that the evidence supports is that capacity sensitivity is a major uncertainty or confounder in the present evidence; the width-23 result for ETHUSD is not evidence that the nonlinear operator is useless but only that this capacity was insufficient for that asset.

Routing experiment. Twelve pre-registered structural states, capacity-matched operators, three seeds. GSPC and BTCUSD prefer the linear operator in all twelve states; ETHUSD reverses across all twelve at once for widths of at least 64, an asset-wide rather than state-specific difference. Across-state spread shrinks with capacity and per-seed directional consistency falls from 31/36 to 28/36. The verdict is that there is no robust sample-level routing evidence.

G. Why "execution completed" is not "hypothesis supported"

Four classes of evidence establish the distinction. First, an insufficient sample cannot support a conclusion: the component-ablation rerun used one seed and two assets and could only be judged partially supported. Second, implementation details can contaminate an ablation: two architecturally identical variants differed by 8.28% because of initialization. Third, an uncontrolled capacity becomes a confounder: a change in capacity alone suffices to reverse the direction of the conclusion. Fourth, an ablation can only speak to "within this architecture," not to necessity in principle, and both the paper and the repository state this limitation explicitly. In addition, the repository rules that where an estimate has a sign but its interval crosses zero, the only permissible formulation is "directionally so, statistically inconclusive," never an upgrade to "supported."

Taken together, the evidence on both sides shows that the value of nonlinear computation is **conditional**, that is, a function of asset, capacity, and market condition; there is no asset on which the nonlinear branch is better always and everywhere.

VII. Experimental Governance: Six Gates and the Release System

A. The four principles

P1, one number, one provenance. Every reported number must resolve to an artifact hash and the configuration that produced it.

P2, assert before executing and score honestly. Each gate is scored 1 (automated and repeatable), 0.5 (human-executed with a documented result), or 0 (absent or failing). A stage's score is the **minimum** over its gate groups, not the mean, so any gap within a stage prevents that stage from passing.

P3, every deviation produces an assertion. A deviation record that ends in a disposition but adds no new check counts as unfinished.

P4, the count of trials is part of the result. N must be declared in advance and reported honestly, with a deflation table for the winner.

B. The six gates

One rule governs the relation between layers: nothing in the upper layers may be used to revise a verdict at L0 or L1.

The incident index I1–I7. Each clause corresponds to a recorded incident: I1, one baseline reaching 15.96 times the parameter count of its reference; I2, a status field contradicting its termination field; I3, an empty root path causing 15 fits to be numbered as one attempt; I4, a silent `nanmedian` dropping 96 windows; I5, OHLC violation rates of 1.1% to 6.4% in five FX feeds left unquantified; I6, a cross-platform newline conversion breaking every frozen hash; I7, a hand-written fixture inconsistent with the frozen configuration.

C. Releases, receipts, and evidence closure

Protocol and attempt state machines. Protocol states are draft, reviewed, approved design, stage released, and closed or superseded. Attempt states are planned, started, completed unreviewed or failed or interrupted, and validated or rejected. The document notes specifically that the count of directories, the count of receipts, and the number of distinct file hashes are none of them the number of independent experiments.

Execution release. See `docs/management/v1_4/AUTHORIZATION.md`. Design approval does not itself permit work on real data. A release must state the decision-maker, the time, the repository and commit identity, the protocol and source hashes, the data contract, the permitted stages and input roots, the output directory, the device and budget, the failure handling, and the revocation conditions. The order is:

TABLE IV
CAPACITY SENSITIVITY: ΔMSE AND THE SHARE PREFERRING NONLINEARITY

Width	Parameters	GSPC	BTCUSD	ETHUSD
23	9 431	$+0.05982$ (0/36)	$+0.17832$ (0/36)	$+0.01454$ (12/36)
64	26 200	$+0.02503$ (7/36)	$+0.18884$ (0/36)	$\mathbf{-0.02807}$ (36/36)
128	52 376	$+0.00606$ (8/36)	$+0.29983$ (0/36)	$\mathbf{-0.03205}$ (36/36)

TABLE V
THE L0–L5 EXPERIMENTAL DOCTRINE AND ITS KEY CLAUSES

Layer	Question answered	Key clauses
L0 Input eligibility	Is the data itself admissible	Content hashes verified at consumption; the time grid must be declared; commensurability across series allowed through exactly one pre-declared mechanism; **silent removal of NaN is prohibited** (removal via a NaN-skipping reducer is still removal); defect handling declared before outcomes
L1 Implementation fidelity	Is the declared baseline actually running	Each model must name its reference implementation and configuration and be diffed clause by clause against it; a capacity-scale check is required; a model deviating from its reference by more than the declared factor "is not that baseline"; L1.8: "a check on a frozen configuration reads the frozen configuration"
L2 Calibration	Does the hardware run it	**This layer produces no scientific claim**; losses are not ranked or published; status and termination fields must not contradict each other; a device change mid-stage is a protocol event
L3 Screen	Broad screening	Evaluation keys are generated before models run; where a feedback loop exists the final evaluation is sealed (empirical loss: 63.7% of entrants beat the benchmark on their best submission against 48.4% on the submission finally chosen, a gap of 15.3 percentage points)
L4 Confirmation	The sole source of primary conclusions	Estimand, strata, and pooling rule declared before outcomes; an active confirmation protocol is not modified, and changes take a new protocol identifier; the trial count is declared rather than discovered; the winner carries a deflation table; selection overfitting is estimated by CSCV with a threshold of 0.05; every gate names who can satisfy it
L5 Temporal robustness	Does it hold in another period	Rolling splits use explicit end-exclusive date boundaries; a fixed-ratio split is never relabelled as rolling-origin evaluation; each split may borrow only the preceding `seq_len` rows; post hoc sensitivity analyses are labelled exploratory

data-eligibility release, preprocessing release, fit release, and evaluation release. Access to real data is currently closed.

Receipts. Each run produces an immutable metadata record whose fields include the logical run identifier, attempt number, attempt identifier, superseded attempt, retry reason and retry authorization, creation time, protocol version, frozen commit, configuration fingerprint, data hash, full command, environment fingerprint, array shapes, finiteness check, diagnostics, and per-file hashes. Reconciliation finds 728 of 729 records mapping uniquely to a plan line, covering 675 distinct plan lines; a further five quarantine copies are byte-identical to their originals and do not count as new attempts; and 699 remain in the completed-unreviewed state.

Retries. A retry creates a new attempt and records what it supersedes and why; existing checkpoints and logs must not be overwritten, and the best run must not be selected while the remaining failures are hidden. It is stated explicitly that a retry is not permitted because of poor performance; only technical failures may be retried, and then only under a separate retry authorization.

Four-fold acceptance. Run completion, artifact receipt, structural acceptance, and scientific usability are four distinct fields. Structural acceptance checks file integrity and identity; numerical acceptance requires separate authorization and checks common keys, ground-truth consistency, scale, metric recomputation, and non-finite outputs; research acceptance then checks pre-registration, test exposure, and claim scope. The document states: "synthetic tests do not demonstrate real generalization, and a hash match does not demonstrate algorithmic correctness."

Evidence closure. The append-only decision ledger numbered 00 to 29 under `reports/evidence_closure/`. Negative results, execution anomalies, and frozen decisions are all retained as evidence.

Dataset rights review. The file `docs/DATASET_RIGHTS_REVIEW_V1_2026-10-10.md` with its 61-row ledger covers 61 OHLC datasets. All rights statuses remain to be verified; every item must be reviewed before data or results are supplied externally.

D. Validation contamination across rolling rounds: a gap at L5

In a rolling evaluation, the test interval of one round becomes the validation interval of the next. Let round one train on 20–21, validate on 21–22, and test on 22–23, and round two train on 21–22, validate on 22–23, and test on 23–24. The interval 22–23 was then observed as a test set in round one and serves as the validation set in round two. The data themselves are not contaminated, but the **researcher's judgement already carries information about that interval**, so the validation set loses its independence and can no longer discharge its two duties: deciding when to stop training and selecting among configurations.

This differs from ordinary data leakage. The latter is the entry of values into training and can be handled by purging and embargoing; the former is the entry of judgement into the process and is undetectable at the data level.

Status in this repository. Clauses L5.1 to L5.4 cover date boundaries, naming discipline, borrowing of input context, and forward-declared sensitivity analyses; *none of them addresses this problem*. The two nearest clauses elsewhere in the doctrine cannot substitute for it either. L3.4, on feedback loops, points in the right direction (the empirical loss it cites is 15.3 percentage points), but its wording is aimed at loops that return scores during development and does not name rolling rounds. The purging and embargoing of L4.9 addresses label-interval overlap within a single split, and the repository has verified that the condition there holds by construction (`Dataset_Custom` sizes each split's usable range as $\text{len} - \text{seq_len} - \text{pred_len} + 1$, so labels never cross the boundary); it therefore handles values, not judgement.

The problem is not hypothetical. Stage D is designed so that, for test year Y, training ends at the end of year $Y-2$, validation uses year $Y-1$, and testing uses year Y, with test years 2023, 2024, and 2025. Substituting, the test year of the 2023 round becomes the validation year of the 2024 round. Stage D stands at 0/144 and has not been run, so the cost of adding the rule before that stage starts is at its lowest. The pre-registration already contains the phrase "the rolling-split implementation gap" (no Stage B–D training is authorized until the Stage A artifacts and that gap have been reviewed), showing that the problem is acknowledged but not narrowed to the visibility of a previous round's test results.

Suggested way to supply it. It would be better to attach the addition to the existing L3.4 and to describe the visibility of test results across rolling rounds as one form of that feedback loop than to create a new clause. The reason is that the basis note of L4.4 states explicitly that presenting a stronger reading as a ported requirement would be inventing a rule rather than porting one.

VIII. The R1 Forward-Increment Track

Content. Protocol `R1-exploratory-small-panel-20261007-v1`. Its principal difference from Phase 1 lies in the forecast target: 96 rows of OHLC history are used to predict the next observed close, that is, a one-step-ahead increment. The four assets are BTCUSD, ETHUSD, GSPC, and N225. The design file is in the local working tree under `research_work/forward_increment/` and is referenced from `docs/management/v1_4/r1_release_candidate.json`.

The three arms. Denoted F, L, and B, with F and L as the primary comparison and B together with persistence (a naive continuation forecast) as auxiliary. The primary gain quantity is $\text{MSE}(L) - \text{MSE}(F)$. The document notes specifically that comparing $F - B$ directly changes several factors at once and therefore "cannot by itself demonstrate a nonlinear mechanism." Ridge regression is included as a candidate solver.

Pipeline and budget. The stages pass in order: source and internal-use basis; custodian quality check; fixed split, row, and key identity; independent review of the exact F/B synthetic lifecycle; T0 interface negative tests; synthetic resource check and executable manifest; fit release; validation selection and freeze; evaluator release; and full stratified reporting. The hard ceiling is CPU, one concurrent fit, two threads, zero data workers, float32 for the neural component and float64 for ridge and aggregation, a cumulative 12 hours, 30 minutes per neural fit, and zero paid spend. The plan calls for 24 neural fits, and 24×30 minutes exactly exhausts the 12 hours; the document states explicitly that no time is left for ridge, evaluation, or aggregation, so the estimate must be redone with synthetic timing before release, failing which a new protocol amendment must be drafted.

Current state. The candidate file is in state `DRAFT_BLOCKED` with `execution_permitted` false and both issuer and issuance time null; all four stages (custodian check, preprocessing, fit, evaluation) are not issued. The stated reason is that administrative delegation exists but the staged operational evidence has not been accepted, and the candidate is not an authorization.

IX. Reproducibility, Naming, and Boundaries

A. Reproducibility versus replication

These two terms carry opposite meanings under different disciplinary conventions, and doctrine clause X7 requires that terminology not be inherited silently. The project's approach is to bind conclusions to protocol, time, and reviewable artifacts rather than to supply a runnable script alone. The README likewise notes that cloning the repository is not sufficient to reproduce an experiment fully, because large training logs, checkpoints, and some prediction artifacts may remain only on the machine where they were produced.

B. Inconsistency between directory names and configuration

`run.py` composes result directory names from arguments such as `args.n_heads`, so every result directory is labelled `dm32_nh8_el2_dl1_df2048`. In `models/derefusion/DeReFusion.py` on the `main` branch, however, the DeReFusion family takes source-level constants: `n_heads` defaults to 4, the Transformer has a fixed one layer, `dim_feedforward` equals $d_{\text{model}} \times 4$, and dropout is 0.3, whereas the pre-registration declares $8/2/2048/0.1$. The configuration claimed by the directory name, the configuration required by the pre-registration, and the configuration the code actually runs are therefore three different things. Clause L1.8, "a check on a frozen configuration reads the frozen configuration," corresponds exactly to this class of problem. Because it undercuts the premise that both sides run with identical hyperparameters, the point should be raised proactively in any external account.

C. The Navier–Stokes boundary

The repository contains an idea that takes the equations of fluid mechanics as architectural inspiration. The governing rules state that this idea has neither been verified nor been refuted within this repository. The sources are Sections 2.3 and 4 of `docs/LITERATURE_AND_NEXT_PLAN.md`, `reports/ns_hypothesis_charter.md`, and Section 6.3 of the README. The substance is that the OHLC panel has no corresponding state variables, invariants, boundary conditions, or known partial-differential-equation residual, so "NS-inspired" may not be upgraded to "physics-informed" unless measurable state, invariant, and falsifiable differential constraints can be supplied independently. The prescribed wording for failure is that "the tested operator inductive bias did not improve the matched forecasting benchmark," and it is stated explicitly that this must never be generalized to "Navier–Stokes is false in finance."

D. Retired phrasing and the publication boundary

Retired phrasing. The phrase "representation bottleneck" is retired; at most "capacity artefact" should be used sparingly, because changing the width alters representational power, optimization, regularization, variance, and implicit bias together. Separately, a feature–outcome relation within the same segment must be described as "contemporaneous and descriptive" and may not be called "leakage."

Publication boundary. The repository states that this is not blanket clearance for the whole repository. Before an experiment bundle is published, its data rights, confidential content, raw and derived data, credentials, local paths, checkpoints, and user-authored changes must all be reviewed item by item; restricted datasets must stay outside version control, with only acquisition instructions, schemas, version identifiers, checksums, and transformation code published.

X. Coverage

This report consolidates and rewrites two earlier documents, for a total of 82 terminology entries. The repository's `main` branch tracks 1563 files, comprising 569 `.npy`, 276 `.txt`, 145 `.py`, 135 `.json`, 125 `.md`, 112 `.csv`, 63 `.tsx`, 48 `.pth`, and 35 `.log` files. Data and weight files are excluded from the concept denominator so as not to inflate the figures.

TABLE VI
Coverage statistics

Basis	Numerator and denominator	Share
Concept coverage	82 entries (about 79 distinct concepts) / 78 concepts enumerated for explanation (self-compiled)	close to 100%
File anchoring	about 75 repository files and directories / 270 (145 `.py` plus 125 `.md`)	about 28%
Document reading	about 32 documents / 44 `.md` files under `docs/`	about 73%

The number of entries exceeds the number of concepts because some entries cover several concepts and a few concepts are described twice from different angles. The denominator of 78 is this report's own enumeration, not an official count by the repository.

Four classes of material remain uncovered: the original publication source of each opponent model; a line-by-line account of the analysis scripts under `reproduction/analysis/`; the implementation details of the data loaders under `data_provider/`; the front-end project `research-ops-site/`; and the field-level structure of the two large JSON files under `docs/management/`.

Note

This report is compiled from commit `87bc113` (2026-10-13) of the repository's `main` branch, with paper figures taken from the original PDF; should the repository be updated, its most recent state governs. Items marked as gaps or inconsistencies are observations drawn from the repository documentation and are offered for discussion.

References

[1] C.-C. Hsieh and M.-Y. Chen, "DeReFusion: A controlled comparison of soft computing fusion strategies for financial time series forecasting via a decomposition-residual architecture," *Applied Soft Computing*, vol. 203, art. 116252, 2026, DOI: 10.1016/j.asoc.2026.116252.

[2] THUML, "Time-Series-Library," GitHub repository. [Online]. Available: https://github.com/thuml/Time-Series-Library